



राष्ट्रीय भू-सूचना विज्ञान
एवं प्रौद्योगिकी संस्थान
भारतीय सर्वेक्षण विभाग
विज्ञान और प्रौद्योगिकी विभाग

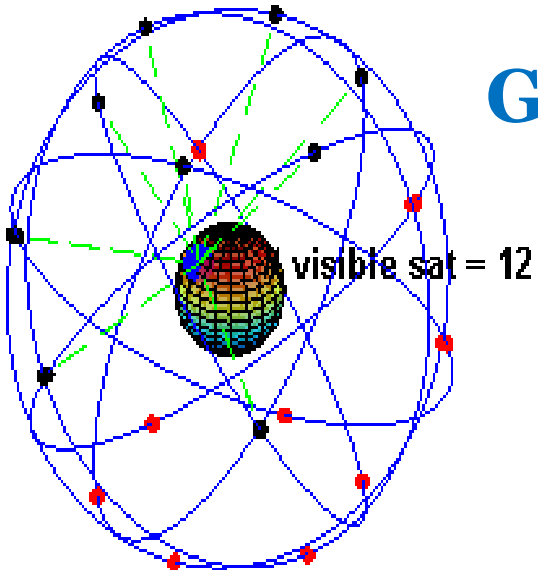
National Institute for Geo-Informatics
Science & Technology
Survey of India
Department of Science & Technology

Route Planning & Transmission Line Survey Using Modern Technologies

Day 11: 20 Jan 2026

GNSS Error Sources & Accuracy Concepts

14:00-15:30



Faculty of Geodesy, NIGST, Survey of India

geodesy.nigst@surveyofindia.gov.in

What to do with Errors?

There are essentially 4 options:

Ignore them

Works if the errors are small (negligible)

Model them

Need good models

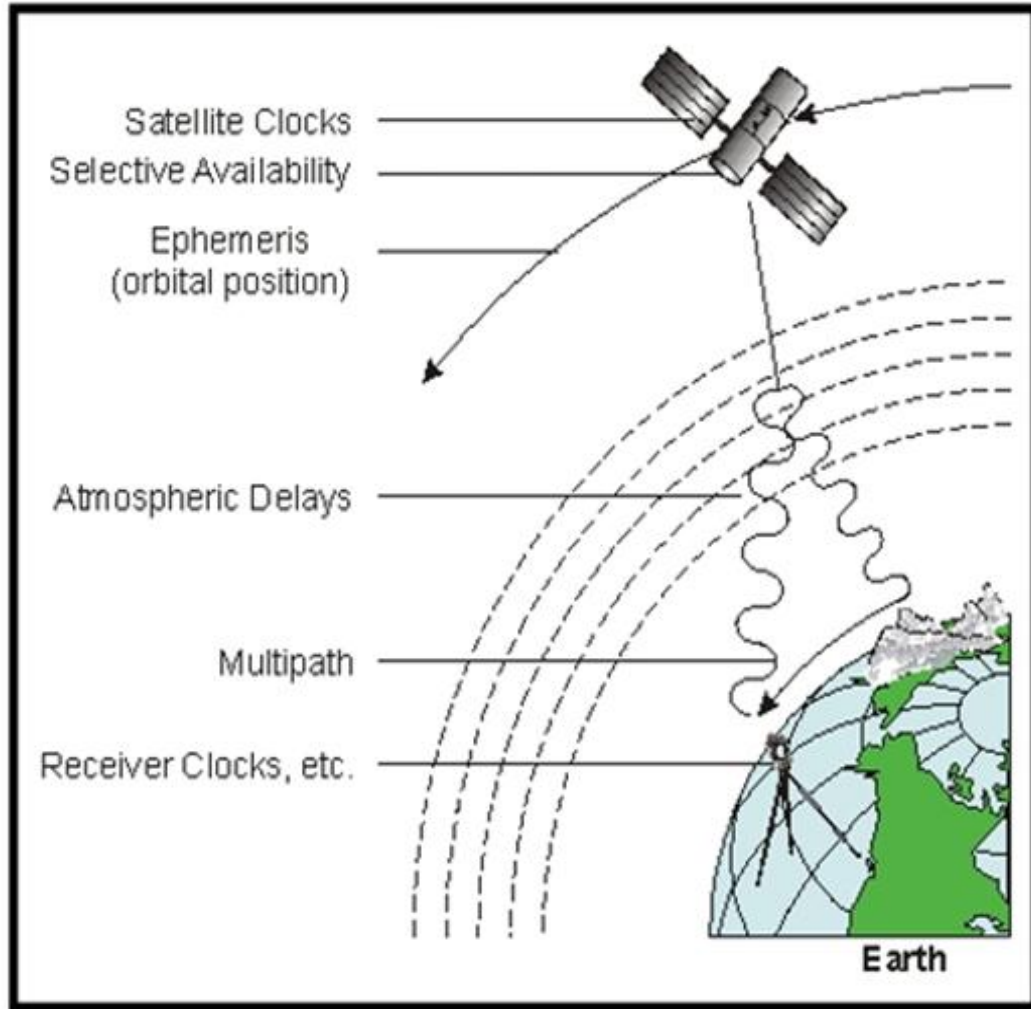
Not all effects can be modeled

Solve for them

Increases complexity of solution

Remove them.

GNSS Error sources and their impact



~ Error Budget (m)		
Error Source	Single Frequency	Dual Frequency
Ephemeris	0.8	0.8
Clock errors	1.0	1.0
Ionosphere	7.0	0.2
Troposphere	0.2	0.2
Multipath	0.3	0.3
Receiver noise	0.1	0.1

Source: <http://www.wirelessdictionary.com>

GNSS ERROR BUDGET

Pseudorange:

$$p = \rho + dp + c(dt - dT) + d_{ion} + d_{trop} + \varepsilon_{mp} + \varepsilon_p$$

ρ (rho) = ρ is the true geometric distance between satellite and receiver.

dp = satellite orbital error — if the ephemeris is not perfect, this term increases.”

$c(dt - dT)$ = clock error component. dt is the receiver clock error, dT is the satellite clock error. The difference multiplied by the speed of light gives meters of error.

d_{ion} = Ionospheric delay. This is the biggest error for single-frequency users.

d_{trop} = Tropospheric delay.

ε_{mp} = term represents multipath error.

ε_p = represents receiver noise.

GNSS ERROR BUDGET (Carrier phase)

$$\Phi = \rho + d\rho + c(dt-dT) + N\lambda - d_{ion} + d_{trop} + \varepsilon_{m\Phi} + \varepsilon_{\Phi}$$

$\rho + d\rho + c(dt-dT)$ The first three terms are same as the pseudorange: geometric distance, orbital error, and clock error.

$N\lambda$ (ambiguity) This is the most important difference. Carrier phase includes an extra term $N\lambda$. N is the integer ambiguity — the unknown number of whole wavelengths. λ is the wavelength of the signal.

Resolving this integer N is NECESSARY IN high-precision GNSS.

$-d_{ion}$ In the carrier phase, the ionosphere appears with a negative sign. This is due to the way the phase advances inside the ionosphere.

d_{trop} Tropospheric delay remains positive, just like in pseudorange.

$\varepsilon_{mp}\Phi, \varepsilon_{\Phi}$ These are multipath and noise terms for the carrier phase.

They are very small compared to pseudorange noise.”

GPS Ephemeris Errors

3 types of ephemerides

- Almanac – very crude ($\sim 100\text{m}$), used only for planning purposes
- Broadcast – reasonably accurate ($\sim 1\text{m}$), used for real-time work
- Precise – very accurate ($\sim 10\text{cm}$), used for high precision work

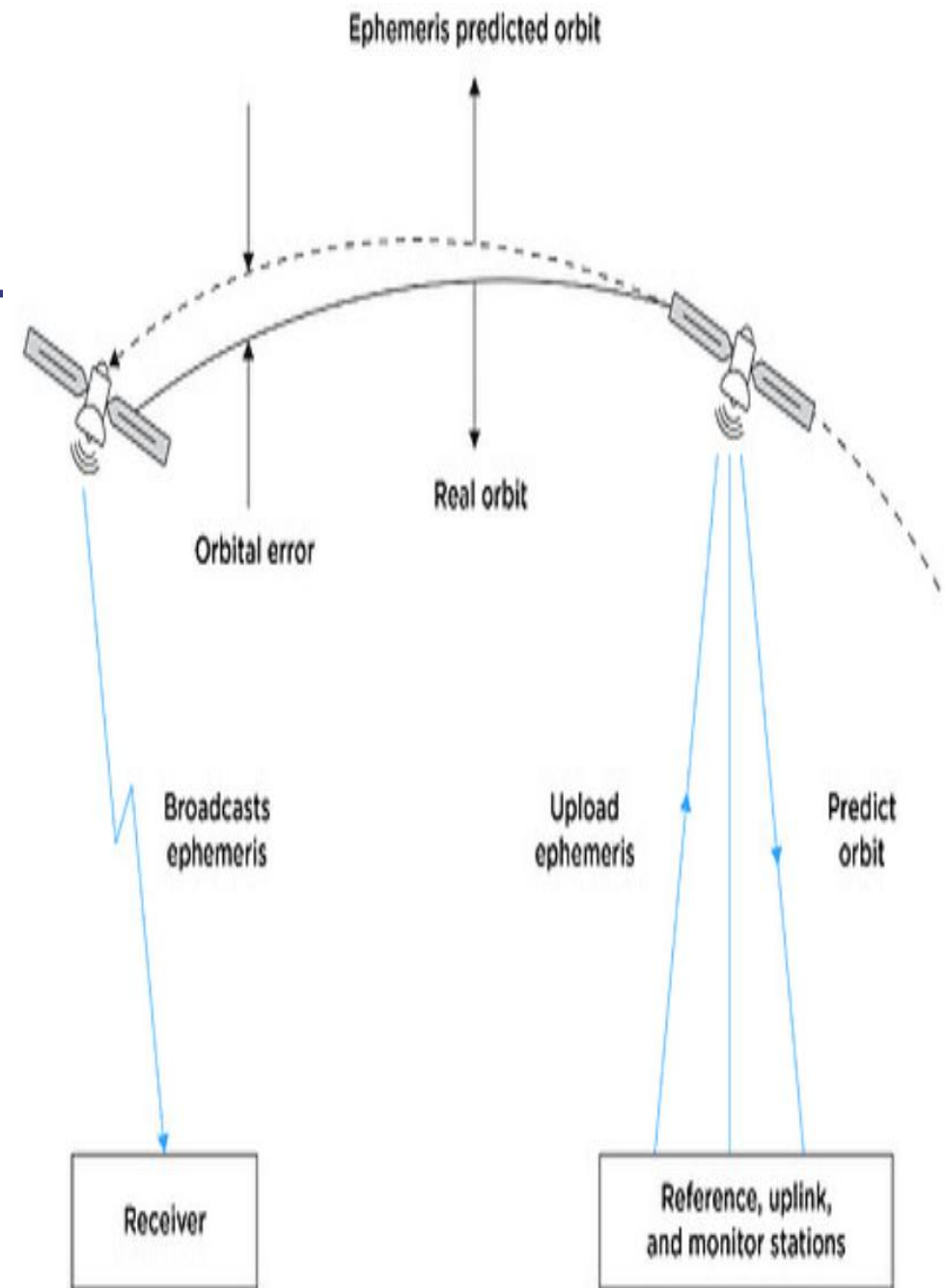
Available on weekly basis approximately 13 days after the end of the solution week.

Ultra-Rapid available 4 times a day almost on near realtime basis

Rapid orbits available 17 hrs after day of observation (UTC)

EPHEMERIS ERRORS

- Ephemeris data includes the satellite's orbital parameters used to calculate its position in space.
- Causes of ephemeris errors include:
 1. Gravitational disturbances
 2. Solar radiation pressure
 3. Imperfect orbital modeling
- These errors lead to discrepancies between the satellite's actual and predicted positions of a few meters.
- *Because every GNSS user observing the same satellite sees the same ephemeris error, differencing measurements between receivers close together can largely cancel this error out, improving relative positioning accuracy;*



SATELLITE CLOCK BIAS (dt)

- GPS satellites use atomic clocks
- Drift allowed = up to 1 millisecond
- Relativity causes:
 -

Ground control stations constantly monitor Satellite clock performance and upload clock correction data via navigation messages to mitigate these errors.



Cesium Frequency Standard (Clock)
Source: [National Air and Space Museum, Smithsonian Institution](#). [1]

RECEIVER CLOCK BIAS (dT)

- Receivers use quartz oscillators
 - Much less stable → large time error
 - Causes ≈ 1500 m position error if unsolved
 - 4th satellite needed to solve receiver clock offset
- The third-largest error, which can be caused by the receiver clock, is its oscillator. Both a receiver's measurement of phase differences and its generation of replica codes depend on the reliability of this internal frequency standard.

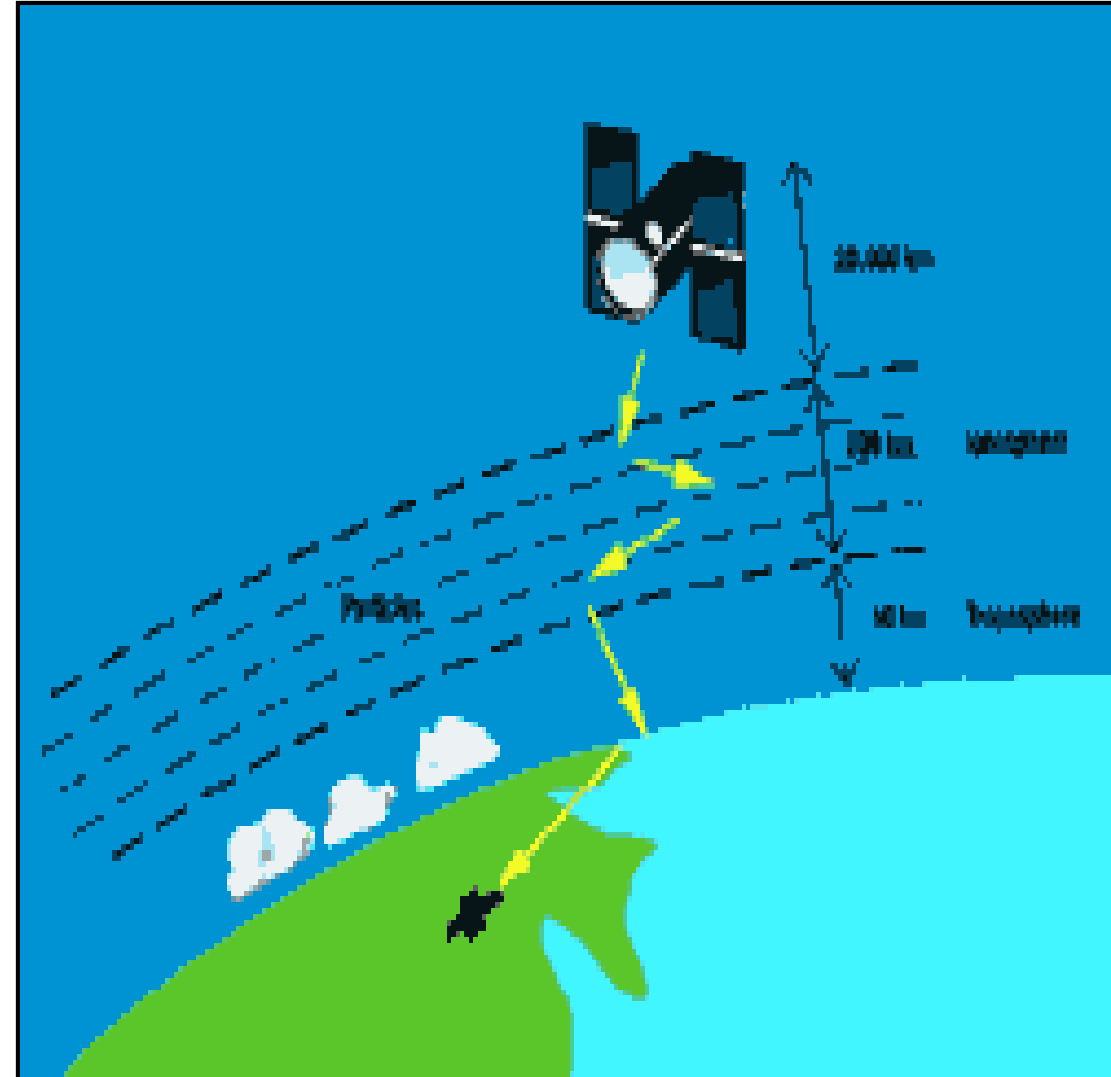
With four satellite signals, the GNSS receiver can simultaneously solve for its position and the clock bias. This approach transforms the time bias into a solvable parameter within the least squares adjustment or positioning algorithm

The Receiver Clock Bias, dT



Earth's atmosphere

- Calculate the Range to a satellite by multiplying a signal's travel time by the speed of light.
- But the speed of light is only constant in a vacuum... Speed varies while travelling through atmosphere.
- Earth's atmosphere is heterogeneous
- Can cause signals to slow down or speed up
- Eliminated by 'dual frequency' receivers (Ionospheric error part)
- Low frequency affected more than high frequency
- Receiver evaluates signal and corrects for error.



NATURE & VARIABILITY OF THE IONOSPHERE

(d_{ion})

1. Ionosphere as Ionized Plasma

Contains free electrons and ions

GNSS signals undergo refraction and diffraction

Delay depends on electron density

Higher TEC $\rightarrow$ higher ionospheric error

2. Ionospheric Variability

Daily: lowest near midnight, highest at noon

Seasonal: worst near March equinox

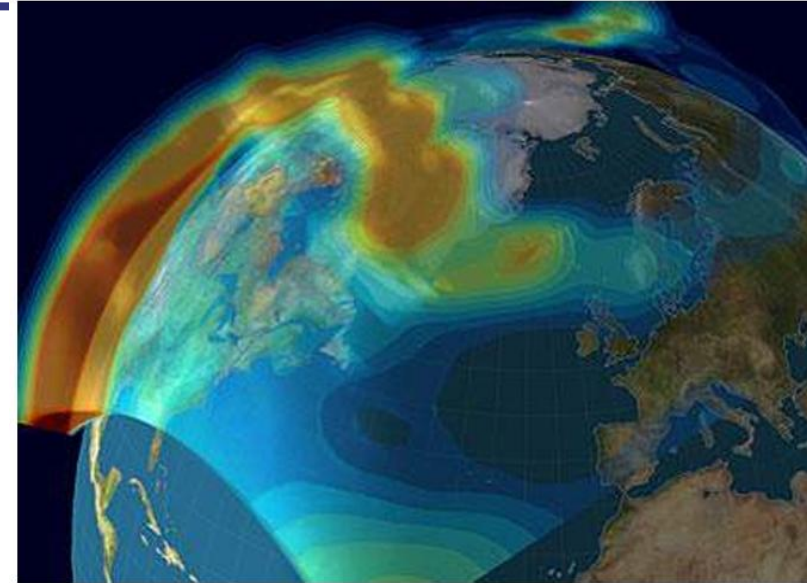
Solar Cycle: increases during solar maximum (11-year cycle)

Geographical: highest TEC around $\pm 30^\circ$ magnetic latitude

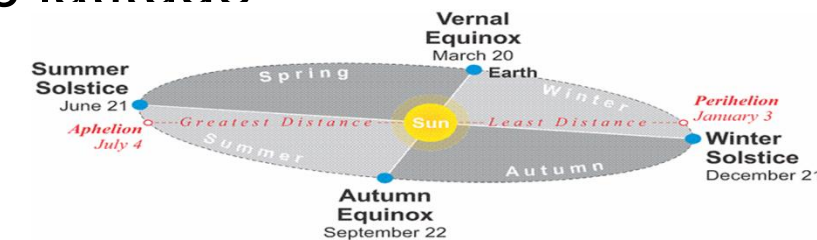
3. Key Parameter: TEC

TEC controls how much the signal is delayed

TEC is dynamic $\rightarrow$ ionospheric error is dynamic



Imaging the Earth's ionosphere
Source: Cathryn Mitchell, University of Bath



Earth's Orbit
Source: GPS for Land Surveyors

Ionosphere and the Sun

DISPERSIVE BEHAVIOR OF THE IONOSPHERE

1. Ionosphere is Dispersive

Delay depends on signal frequency

Lower frequency → more delay

Higher frequency → less delay

2. Different Effect on Code and Carrier

Code signals slow down → Group Delay (positive error)

Carrier phase speeds up → Phase Advance (negative error)

Code range becomes too long

Carrier range becomes too short

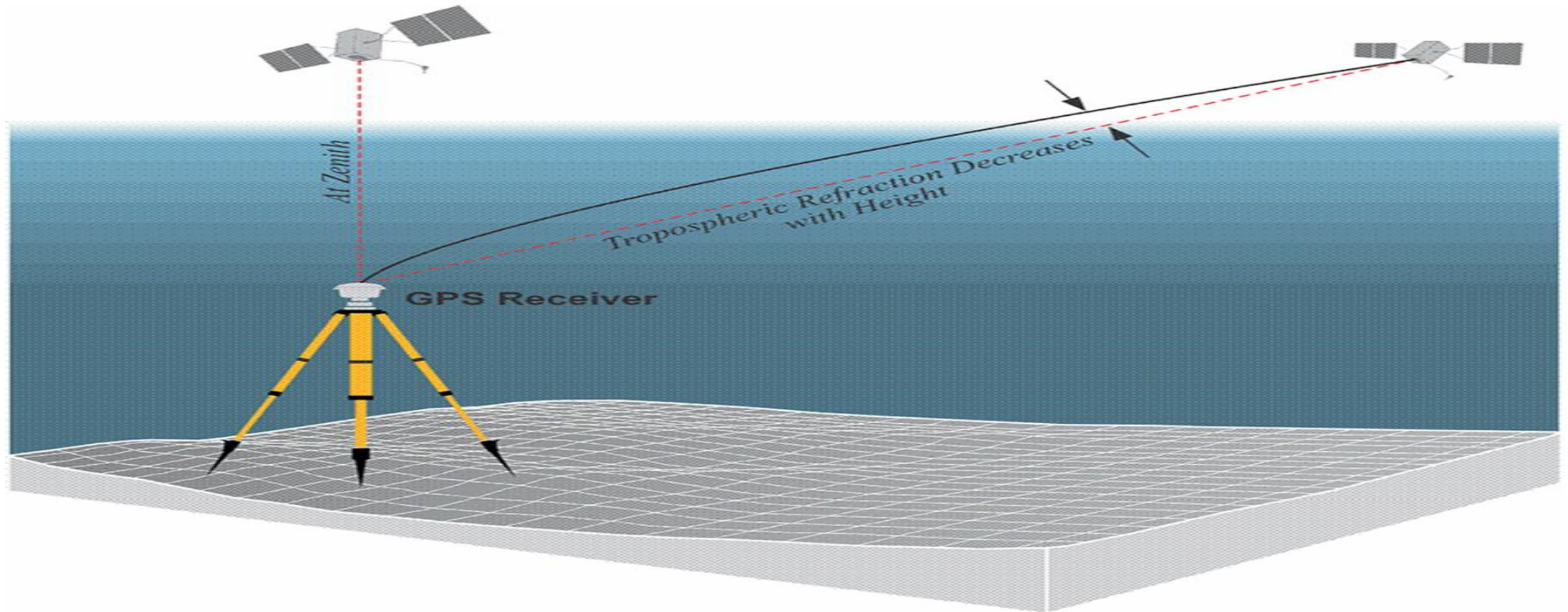
3. Why This Matters

Difference between delays at two frequencies

→ allows dual-frequency receivers to estimate ionospheric delay

TROPOSPHERIC DELAY (d_{trop})

GNSS signals slow down due to refraction inside the troposphere.



Tropospheric Delay

- Caused by the neutral atmosphere, which is a nondispersive medium (as far as GPS is concerned)
- Troposphere extends up to 40 km
- Effects carrier phase and code ranges the same
- Typically separate the effect into::Dry component; Wet component
- Dry component contributes 90% of the error
- Easily modeled
- Wet component contributes 10% of the error
- Difficult to model because you need to know the amount of water vapor along the entire path

TROPOSPHERIC DELAY — NATURE & BEHAVIOR(d_{trop})

1. What is Tropospheric Delay

Extra travel time through the neutral atmosphere

Caused mainly by dry gases + water vapor

Delay always positive (slows the signal)

2. Elevation Angle Dependence

Maximum delay → satellite near horizon

(signal passes through long atmospheric path)

Minimum delay → satellite at zenith

(shortest atmospheric path)

3. Zenith Delay & Slant Delay

Delay is modeled first at zenith

Then scaled to slant path

COMPONENTS OF TROPOSPHERIC DELAY (d_{trop})

A. Dry Component (Hydrostatic Delay)

Contributes 80–90% of total delay

Related to atmospheric pressure

Very stable and predictable

Easy to model (Saastamoinen, Hopfield)

B. Wet Component (Water Vapor Delay)

Caused by atmospheric water vapor

Highly variable and unpredictable

Depends on humidity, temperature, weather

Hardest part to model accurately

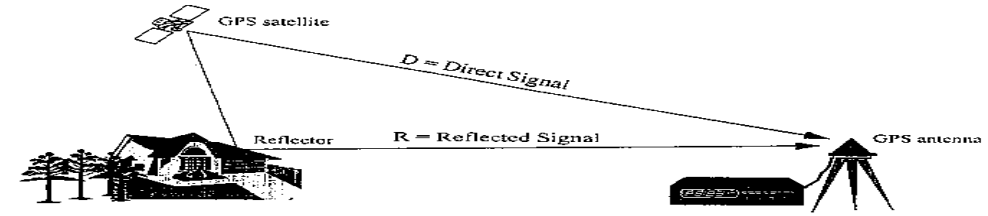
3. Variation with Height

Refractivity **decreases** with altitude

Near surface → higher density → stronger delay

Higher altitude → thinner air → weaker delay

MULTIPATH ERROR



1. What is Multipath

- GNSS signals bounce off reflective surfaces, and reach the receiver after reflection
- Common reflectors : buildings, ground surface, water bodies, vehicle metal roofs, tree trunks
- Both direct + reflected signals enter the antenna

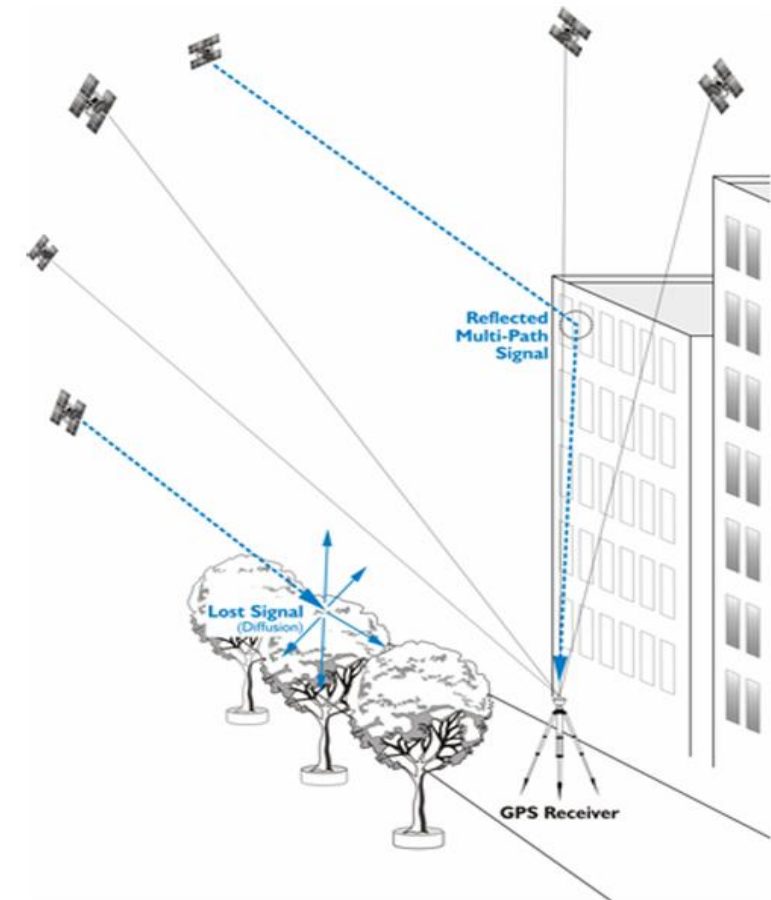
2. Consequences

- Erroneous pseudorange
- Incorrect position
- Distorts code and carrier measurements
- One of the most common field errors in GNSS

3. The reflected signal:

- Arrives slightly later
- Has different phase
- Has reduced power
- Has distorted correlation shape

Multipath, ϵ_{mp} and $\epsilon_{m\phi}$



Multipath
Source: GPS for Land Surveyors

MULTIPATH ANTENNA BASED MITIGATION TECHNIQUE

- Choke ring antennas are the most effective for blocking low-angle reflections. They prevent signals reflected from the ground and nearby objects from entering the antenna.
- Ground planes reduce ground reflections. They create a physical barrier that absorbs or deflects the reflected signals coming from below.
- Placing the antenna away from buildings, walls, vehicles, and metal structures is extremely important. Most multipath comes from the surroundings
- We can set a cutoff angle in the receiver to ignore low-elevation satellites. Low-angle signals travel through long paths and produce strong multipath.

Choke Ring Antenna



GNSS Antenna Ground Plane

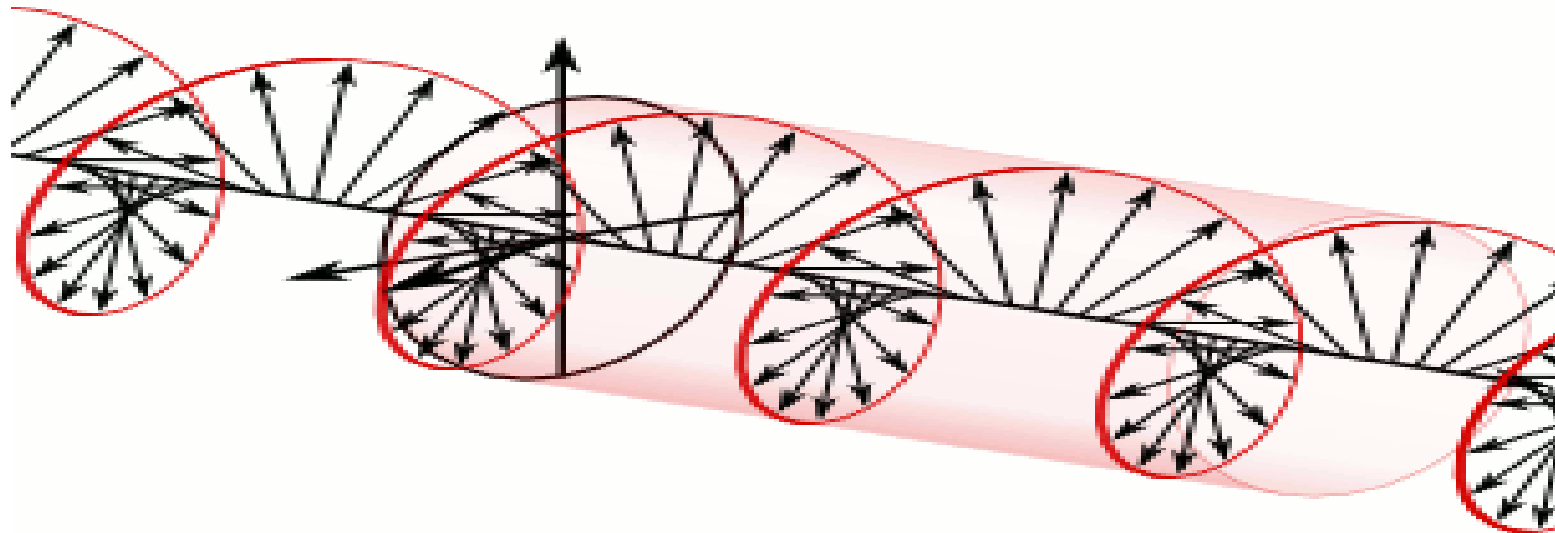


MULTIPATH MITIGATION THROUGH BEST PRACTICES TECHNIQUE

- i. Avoid Problematic Locations “We should avoid setting up GNSS stations near buildings, water bodies, and trees
- ii. Use stable mounting points so that the antenna remains steady during observations
- iii. Prefer open-sky environment - The best mitigation is always a good open environment. Open sky means fewer reflectors

Multipath Errors

- GNSS signal is Right Hand Circularly Polarized (RHCP)
 - After reflection:Polarization becomes Left Hand Circularly (LHCP)
Good GNSS antennas: Prefer RHCP signals, Naturally reject LHCP signals
- This is another antenna-based protection against multipath



EFFECT OF RECEIVER NOISE — CODE VS CARRIER (ε_p , ε_ϕ)

1. Noise Effect on Code (Pseudorange)

Causes **large error** (centimeter level)

Code measurements depend on **chip width** → low resolution

Code tracking is more sensitive to noise

2. Noise Effect on Carrier Phase

Effect is **very small** (sub-mm to mm level)

Carrier depends on **wavelength** (~19 cm for L1) → high resolution

Extremely precise, less affected by noise

3. Key Point

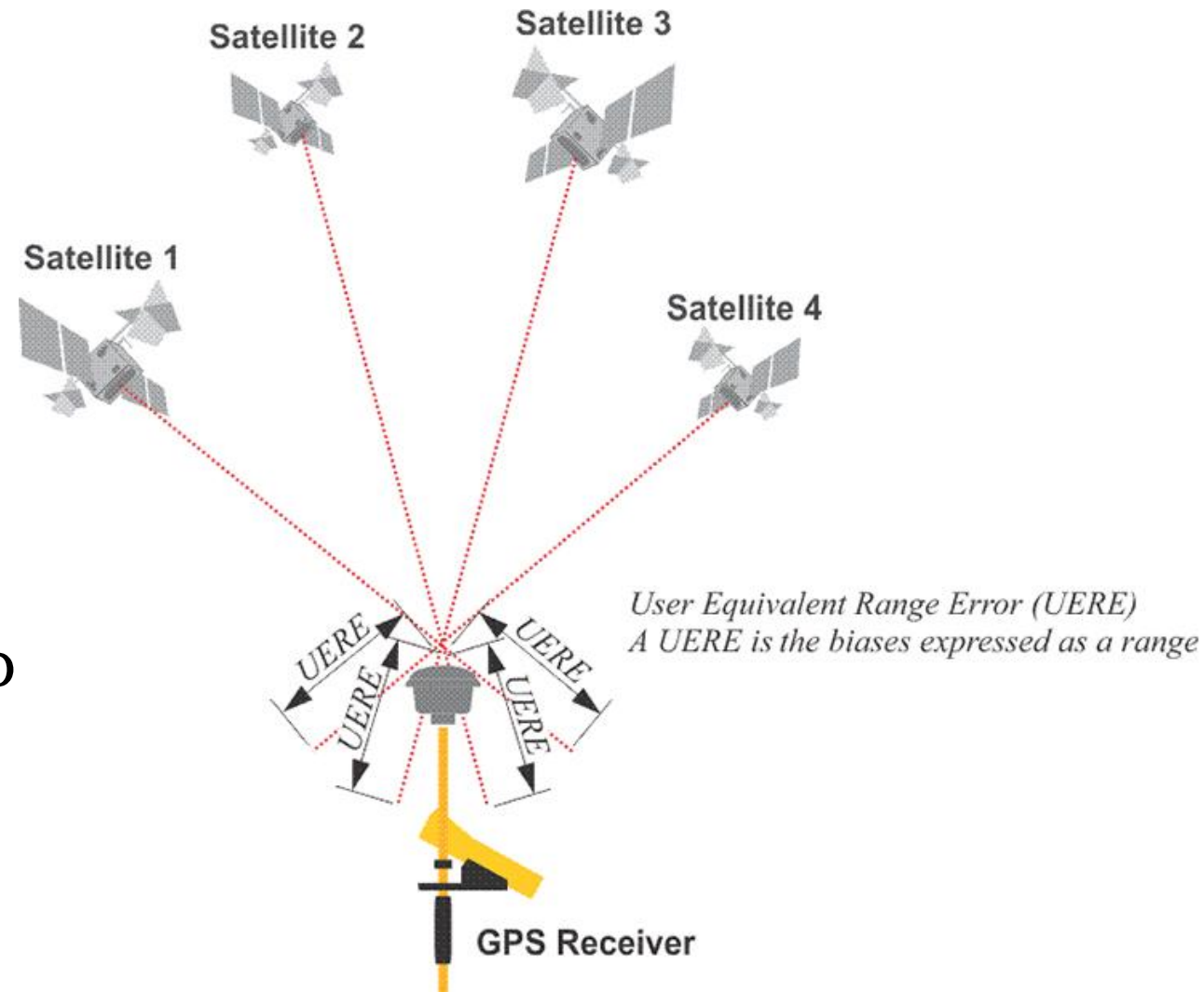
Receiver noise affects code much more than carrier

Carrier phase remains highly accurate even under moderate noise

User Equivalent Range Error (UERE)

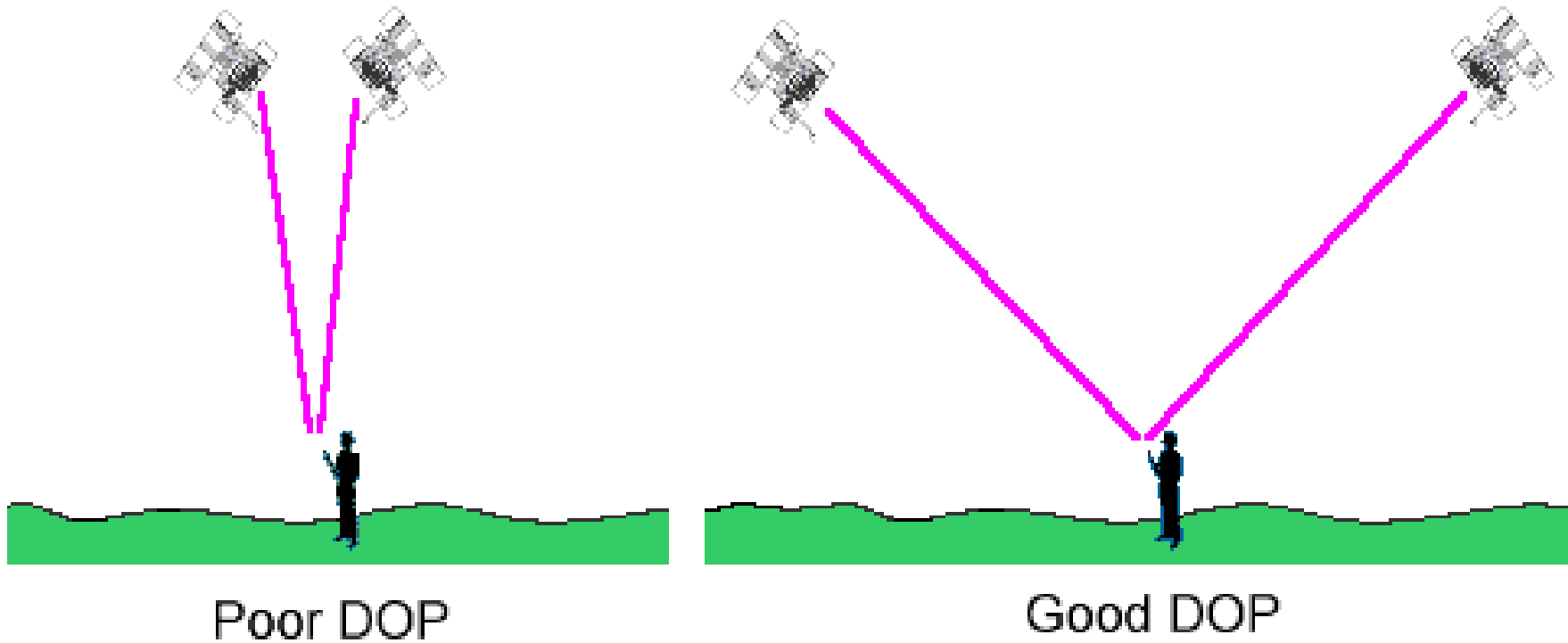
Crude estimate of the expected error
Consists of contributions from

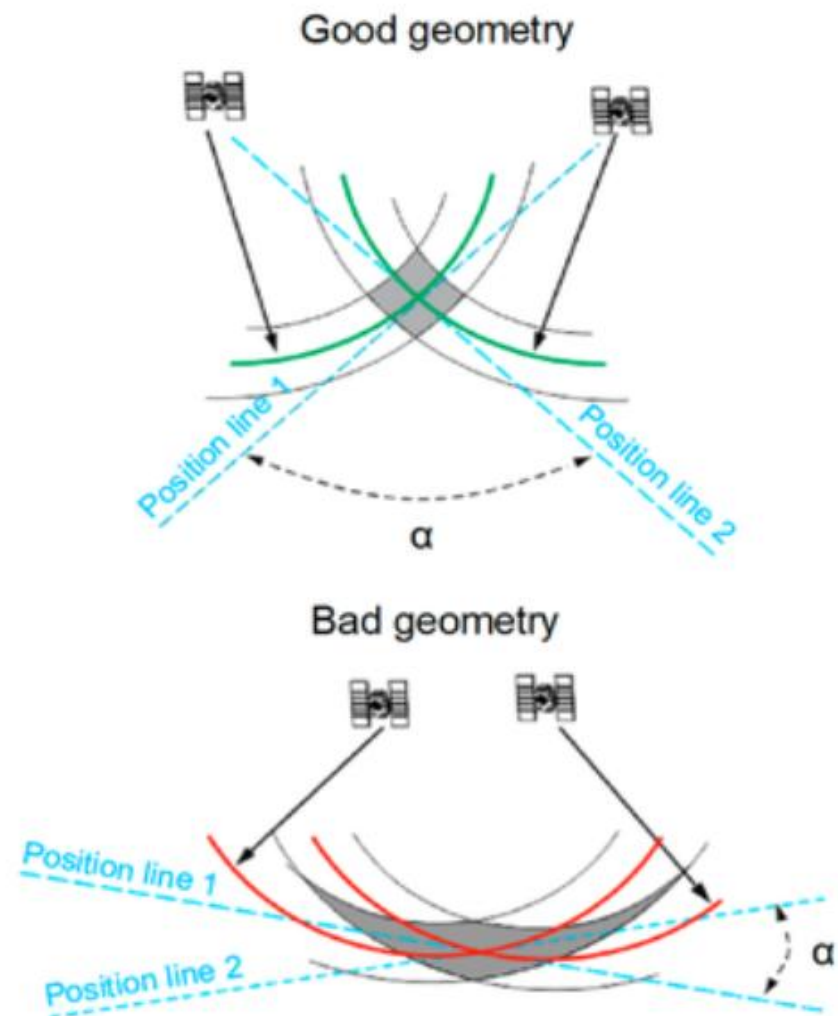
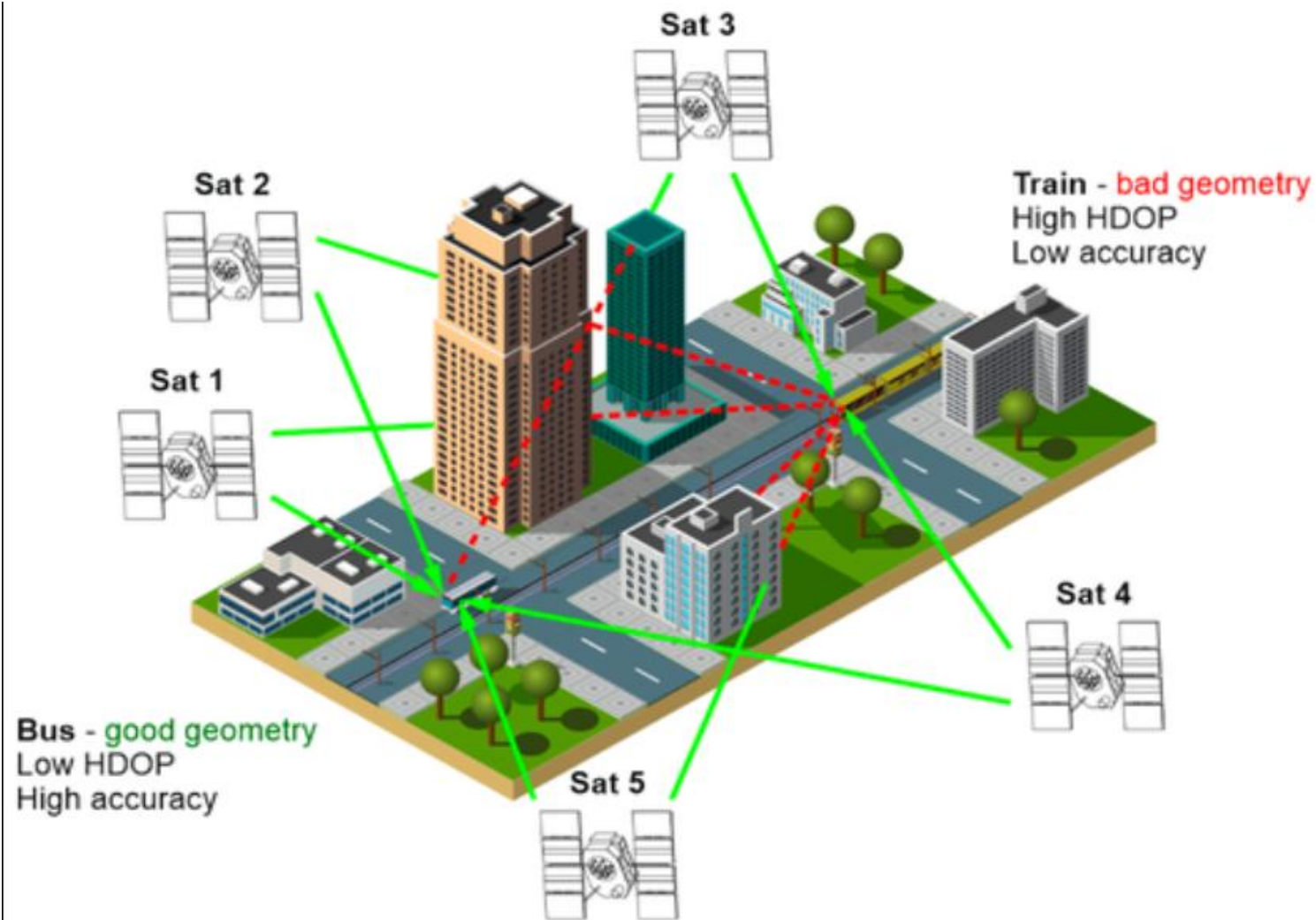
- Measurement noise
- Satellite biases
- Wave propagation errors
- Transmitted through the Navigation message
- Combined with DOP informatio



Satellite geometry

Quantified by DOP: Dilution of Precision





Types of Geometric Factors

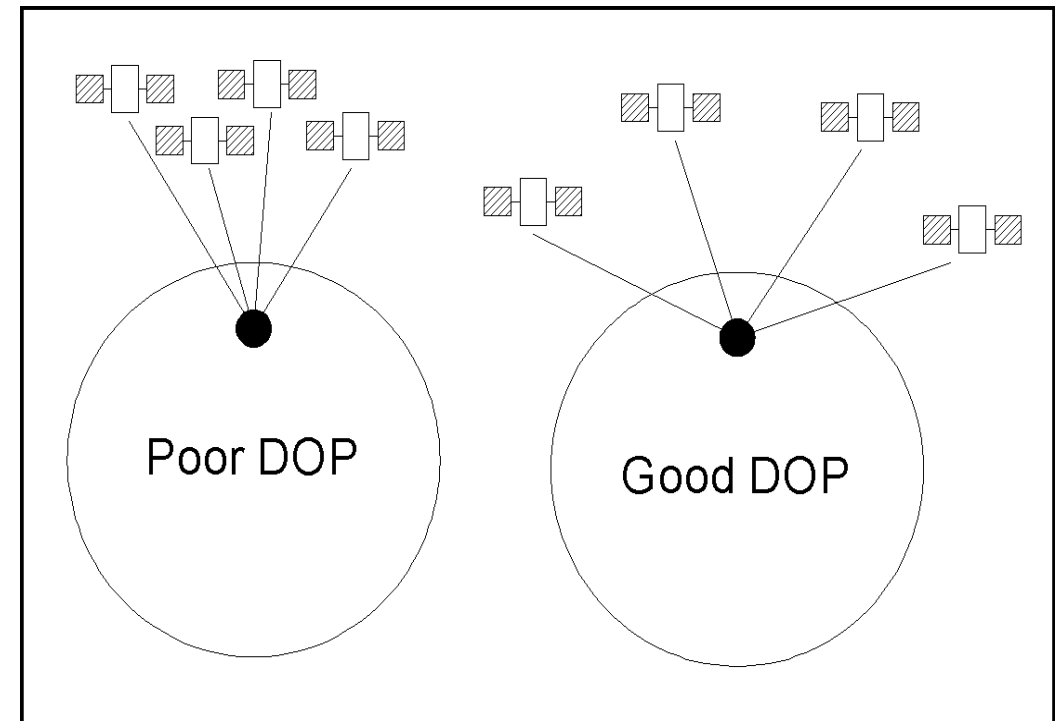
The strength of figure of the satellites is taken into consideration by the dilution of precision (DOP) factor

Depends on number of satellites

Depends on location of satellites

Good receivers determine which satellites will give the lowest GDOP

- Different kinds of DOPs
 - HDOP (horizontal)
 - VDOP (vertical)
 - PDOP (position) (3-D component)
 - TDOP (time)
 - GDOP (geometric) (PDOP and TDOP)



GNSS Augmentation Systems

DOP quantifies the effect of satellite geometry on positioning accuracy.

It does not measure error itself

It amplifies or dilutes measurement errors (UERE) $\boxed{\text{Position Error} = UERE \times DOP}$

DOP depends on: Number of satellites; Spatial distribution of satellites

GDOP – Geometric Dilution of Precision

Position (PDOP); Time (TDOP) $\boxed{GDOP^2 = PDOP^2 + TDOP^2}$

1.

2.

GNSS Augmentation Systems

1.

2.

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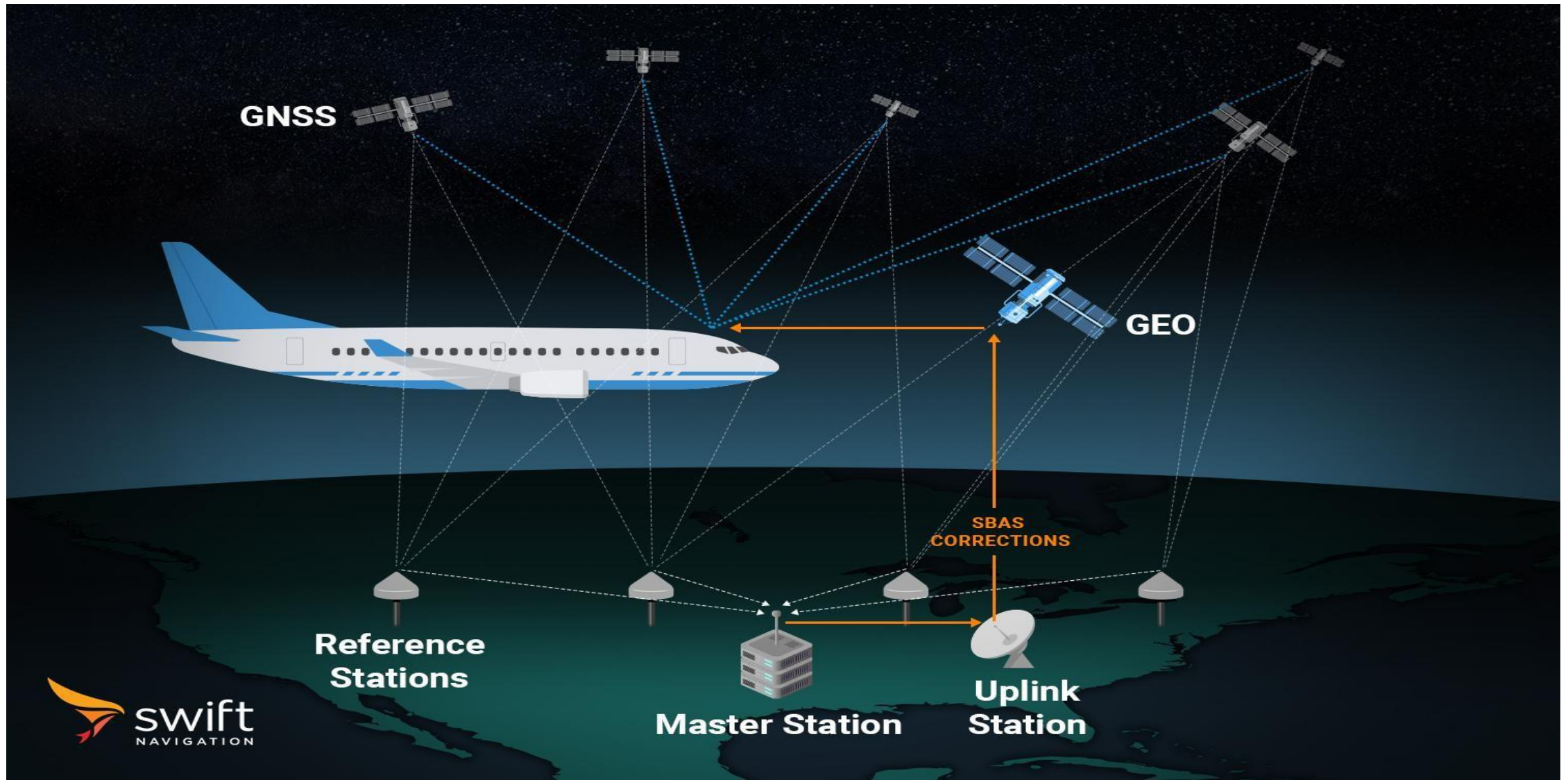
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How to resolve the errors practically?

Contributing Source	Resolving
Satellite Clock	DGNSS/RTK/PPP
Orbit errors	DGNSS/RTK/PPP
Ionospheric delays	DGNSS/RTK/ Dual Frequency/ Modelling
Tropospheric delays	DGNSS/RTK/ Modelling
Receiver Noise	Receiver electronics
Multipath	Short delay in Antenna & Receiver Long delay

SBAS in Aviation

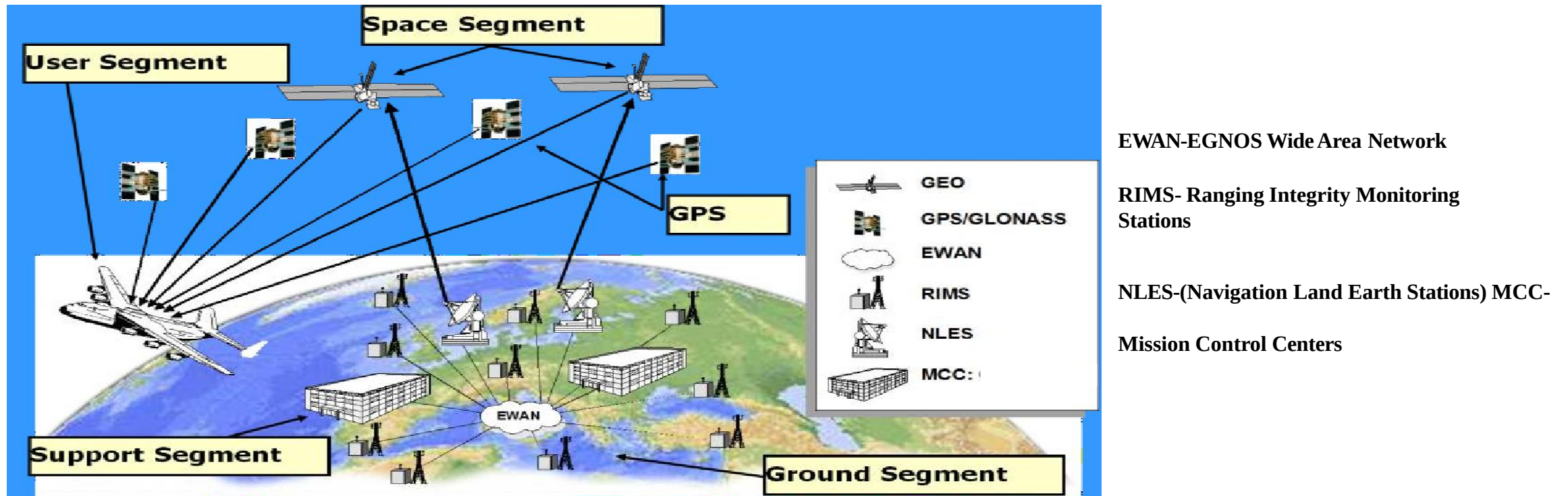


Major SBAS Systems:

1. WAAS (Wide Area Augmentation System) – Operated by the USA for North America.
2. EGNOS (European Geostationary Navigation Overlay Service) – Operated by the European Union.
3. MSAS (MTSAT Satellite Augmentation System) – Operated by Japan.
4. GAGAN (GPS Aided GEO Augmented Navigation) – Developed by India.
5. SDCM (System for Differential Corrections and Monitoring) – Russia's SBAS system.
6. BDSBAS (BeiDou SBAS) – Developed by China as part of the BeiDou system.

Satellite based Augmentation Systems (SBAS)

SBAS is a wide area or regional augmentation system that provides users additional satellite-based broadcast messages.



European Geostationary Navigation Overlay Service (EGNOS) SBAS Architecture

GPS-aided geo-augmented navigation (GAGAN)

The **GPS Aided Geo Augmented Navigation** or GPS and geo- augmented navigation system (GAGAN) is an implementation of a regional [satellite-based augmentation system](#) (SBAS) by the [Indian government](#). It is a system to improve the accuracy of a [GNSS receiver](#) by providing reference signals.

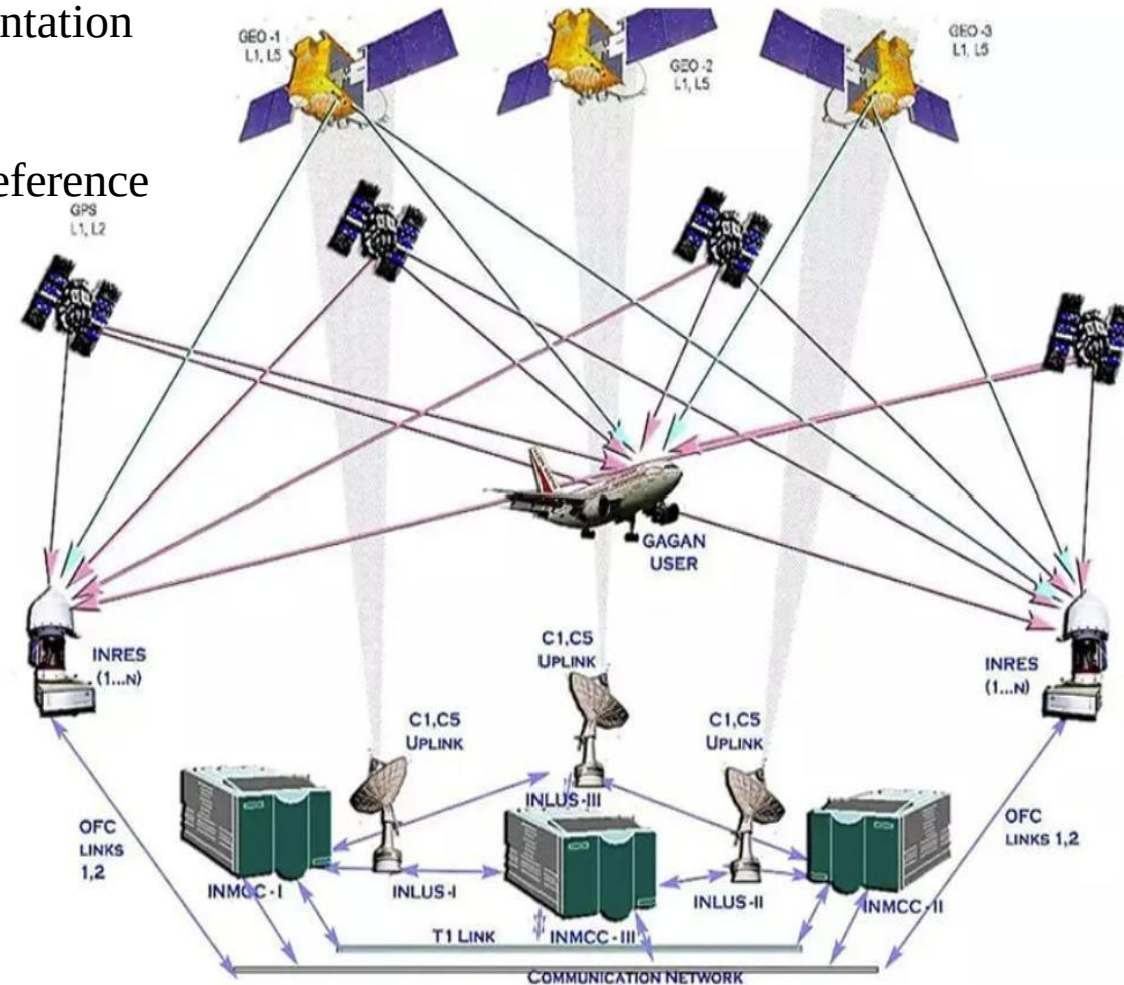
The system uses eight reference stations located in [Delhi](#), [Guwahati](#), [Kolkata](#), [Ahmedabad](#), [Thiruvananthapuram](#), [Bangalore](#), [Jammu](#) and [Port Blair](#), and a master control center at [Bangalore](#).

Gsat-8, Gsat 10 has GAGAN payload

GAGAN is one of the most advanced air navigation systems in the world.

Architecture of GAGAN

- GPS Aided Geo Augmented Navigation is a Satellite Based Augmentation System
- Utilizes a network of ground reference stations known as Indian Reference Stations (INRES) to collect GPS satellite data.
- Data from these reference stations is gathered at the Indian Master Control Centre (INMCC), where GPS correction messages are generated.
- Corrected differential messages are transmitted via the Indian Uplink Station (INLUS) and broadcasted through three geostationary satellites (GSAT-8, GSAT-10, and GSAT-15).
- The information transmitted is compatible with the standard GPS signal structure, allowing any SBAS-enabled GPS receiver to interpret it.
- Used primarily for accurate aircraft landing.



ACCURACY IMPROVEMENT TECHNIQUES

- Dual/triple frequency
- RTK / NRTK
- Good antenna
- Mask angle
- DGPS
- PPP
- Longer occupation
- Low PDOP planning

PRACTICAL FIELD RECOMMENDATIONS

- Open sky visibility
- Avoid reflective surfaces
- Calibrated antennas
- Maintain antenna orientation
- Use multiple constellations
- Observe during low PDOP
- Use precise ephemeris

Thank You